

SYNTHETIC DEMO PROCEDURE · NOT FOR FIELD USE

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Inverter Cooling-System Inspection

Manatee Solar Energy Center · Field Procedure

PROCEDURE	MAN-OPS-INV-014
VERSION	2.1
EFFECTIVE	June 15, 2026
STATUS	Approved — Synthetic
APPLIES TO	HPS-2500X inverter fleet

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4. Response to B17 Thermal Derating

Use this procedure when an HPS-2500X inverter remains online in a B17 derated state. Stop and use the site's higher-severity response if the asset is faulted, offline, damaged, or presents an unsafe condition.

4.1 Preconditions

- Confirm the event is B17 and the current state is Derated—not Thermal Trip.
- Review the ten-minute controller-temperature and active-power summaries.
- Obtain the required work authorization and apply all site safety controls.
- Do not open energized compartments under this synthetic procedure.

4.2 Authorized inspection sequence

- 1 Record controller-temperature peak, configured threshold, ambient-temperature peak, and current active-power output.
- 2 From the exterior, inspect intake screens and exhaust openings for visible obstruction, vegetation, dust loading, or water intrusion.
- 3 Review fan command and fan-feedback indications. If feedback is missing or unstable, place the inverter in maintenance hold and escalate.
- 4 Confirm temperature is declining after derating. If temperature is stable or rising, do not reset; notify the control room and reliability engineering.
- 5 When OEM recovery criteria are met, acknowledge B17 through the approved operator interface and monitor for ten minutes.

Do not reset solely because output is reduced. The protective state must be understood and the cooling path checked first.

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5. Return-to-Service and Closeout

5.1 Return-to-service criteria

CHECK	ACCEPTABLE EVIDENCE	IF NOT MET
Controller temperature	Below OEM recovery temperature and declining/stable for required interval	Remain derated; escalate
Cooling path	No visible obstruction or unsafe condition	Maintenance hold
Fan feedback	Present and stable relative to command	Maintenance hold; fan diagnostic
Post-reset monitoring	No repeated B17 during ten-minute observation	Return to derated state; reliability review

5.2 Required record

- Asset identifier and event timestamp
- Observed temperature, threshold, ambient condition, and output
- Inspection result and any corrective work requested
- Operator acknowledgement time and post-reset condition
- Links to the diagnostic run and cited document versions

Performed by / date

Reviewed by / date